

RESEARCH ARTICLE

Interspecific phenological variation in herbaceous species is more strongly reflected in vegetative traits than in floral traits

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Handling Editor: Jayanti Ray Mukherjee**Abstract**

1. Understanding the role of functional traits in shaping plant phenology is crucial for understanding how organisms adapt to changing environmental conditions. As climate change alters seasonal timing and disrupts ecological interactions, identifying relationships between phenology and traits is essential for predicting species' responses and potential mismatches.
2. While vegetative traits such as plant height and leaf area are often associated with phenological variation, floral traits, which are closely linked to reproductive success, remain largely unexplored, despite their potential to significantly influence phenology. This study aims to understand the relative influence of floral and vegetative traits on patterns of species-specific phenological variation.
3. We monitored the year-round phenology (initial growth, leaf out, onset, peak, end and duration of flowering, fruiting and senescence) of 68 herbaceous perennial plant species in three German botanical gardens (Halle, Jena, Berlin). We measured four vegetative traits—plant height, leaf area, specific leaf area (SLA) and leaf dry matter content (LDMC)—as well as 12 floral traits related to flower morphology, flower number, nectar and pollen, with all measurements conducted for each species in each garden at flowering time. As a basis for the inclusion of floral traits in studies on phenology–trait relationships, we used these data to analyse underlying trait correlations and trade-offs between floral and vegetative traits.
4. Analysis of boosted regression trees that accounted for phylogenetic dependence of the species showed that vegetative traits, especially vegetative height, were most important for explaining patterns in reproductive and vegetative phenology. Taller plants showed later initial growth and later flowering and fruiting compared to smaller plants, and higher LDMC was associated with later senescence. Floral

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